

**Remarks: -**

- 1- All Material are as per ASTM A105 EDITION - 2021 / SA105 OF ASME BPVC SEC-II, PART-A EDITION-2021.
- 2- Visual checked and found free from Cracks, sharp Notches and dent and Mechanical Marks.
- 3- Dimension 100% checked and found acceptable limit as per ASME B16.5 Edition 2020.
- 4- Material is fully killed & fine grain practice.
- 5- Welding has not been performed on Flanges.
- 6- All items meeting general requirement of ASTM 961/ASTM A961M EDITION – 2021 / SA961/SA961M EDITION–2021
- 7- Flange Gasket Finishing on RF Flanges 125 μ in (3.2 μ m) -250 μ in (6.3 μ m).
- 8- Job Hardness checked 2 Nos minimum each Size, per item, per heat treatment lot and found 149 to 165 BHN.
- 9- Material is confirms to NACE MR 0175 /ISO 15156-2015 and NACE MR 0103/ ISO 17945-1:2015.
- 10- All Requirements of material specification and CSA Z245.12-21 Grade 248 Cat-I have been fully complied with.
- 11- **C.E. Formula = $C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$.
- 12- *For each reduction of 0.01% below the specified carbon maximum (0.35%), an increase of 0.06% manganese above the specified Maximum (1.05%) will be Permitted up to 1.35 % For SA105 and 1.65 % For ASTM A105.
- 13- Mechanical Test Sample Preparation as per SA370/ASTM A370 EDITION-2021.
- 14- Melting of origin – INDIA.
- 15- MARKING As per MSS SP-25-2018 :

16	CDI	4" - 900# SCH XS	SA/A105N	HEAT CODE-A2971A	B16.5	NACE	INDIA
22	CDI	2" - 2500# SCH XS	SA/A105N	HEAT CODE-A2972A	B16.5	NACE	INDIA
22	CDI	2" - 2500# SCH XS	SA/A105N	HEAT CODE-A2973A	B16.5	NACE	INDIA
22	CDI	2" - 2500# SCH XS	SA/A105N	HEAT CODE-A2974A	B16.5	NACE	INDIA

Signature /Seal Of Inspector
 (For-Inspection Agency)



Manager (Q.C)
 (C.D. Industries)



MANUFACTURER TEST CERTIFICATE

(According to DIN EN 10204 TYPE 3.1- 2004)

Certificate No : CDI/LAB/2858B Date: 07.11.2023	W.O. No : 2858
Client: M/s. DODSON GLOBAL INC	Inspection Agency : CDI
PO No. 362947-408 DATE 06.09.2023	Order Status : P
Material Specification: ASTM A105 EDITION - 2021 / SA105OF ASME BPVC SEC-II, PART-A. EDITION-2021	

We hereby certify that the material detailed below meets the requirement of purchase order/relevant specification

S. No	Item Description	Material	Offer Qty.	Batch/Lot/ Group No.	Heat Treatment type	H.T. Soaking Temperature	H.T. Soaking Time	Cooling Medium
16	4" - 900# - WNRF - SCH XS	SA/A105N	225	A2971A	Normalized	910°C	2.00 Hrs.	Air
22	2" - 2500# - WNRF - SCH XS	SA/A105N	316	A2972A	Normalized	910°C	2.00 Hrs.	Air
22	2" - 2500# - WNRF - SCH XS	SA/A105N	317	A2973A	Normalized	910°C	2.00 Hrs.	Air
22	2" - 2500# - WNRF - SCH XS	SA/A105N	317	A2974A	Normalized	910°C	2.00 Hrs.	Air
		TOTAL	1175	Nos				

CHEMICAL COMPOSITION (%)

Batch/Lot/ Group No.	Heat/Cast/ Melt No.	Lab Ref No.	C	Mn	P	S	Si	Cr	Ni	Mo	V	Cu	AL	CE**
LADLE ANALYSIS														
	23E00832	15362	0.200	1.08	0.020	0.007	0.280	0.005	0.001	0.001	0.001	0.002	0.050	0.38
	G4/23	15405	0.200	0.980	0.021	0.009	0.280	0.260	0.040	0.020	0.005	0.030	0.022	0.43
	H1/22	15418	0.180	0.910	0.021	0.008	0.210	0.200	0.020	0.020	0.005	0.020	0.031	0.38
	H1/21	15419	0.180	0.950	0.020	0.009	0.230	0.280	0.030	0.020	0.006	0.020	0.022	0.40
PRODUCT ANALYSIS														
A2971A	23E00832	15362	0.207	0.970	0.016	0.002	0.234	0.175	0.025	0.022	0.005	0.020	0.026	0.41
A2972A	G4/23	15405	0.201	0.990	0.021	0.009	0.275	0.258	0.041	0.024	0.014	0.020	0.020	0.43
A2973A	H1/22	15418	0.188	0.968	0.017	0.007	0.266	0.253	0.040	0.024	0.014	0.020	0.020	0.41
A2974A	H1/21	15419	0.195	0.972	0.018	0.009	0.269	0.255	0.040	0.024	0.014	0.020	0.020	0.42
Specification Requirement		Minimum	-	0.60	-	-	0.10	-	-	-	-	-	0.020	-
		Maximum	0.35	1.05*	0.035	0.040	0.35	0.30	0.40	0.12	0.08	0.40	-	0.43

PHYSICAL PROPERTIES

Batch/Lot/ Group No.	Yield Strength (MPa) (0.2% offset)	Tensile Strength (MPa)	Elongation % on GL=4D	Reduction in Area %	Hardness (B.H.N.)	Test Sample Direction	Charpy V- Notch – 10x10x55mm (2mm V- Notch)				Remark
							I	II	III	Average (In Joules)	
A2971A	343	547	31.00	58.91	159, 163	Tangential	-	-	-	-	Satisfactory
A2972A	350	521	31.60	59.04	153, 156	Tangential	-	-	-	-	Satisfactory
A2973A	326	513	31.40	59.04	153, 156	Tangential	-	-	-	-	Satisfactory
A2974A	335	531	32.00	60.94	153, 156	Tangential	-	-	-	-	Satisfactory
Specification Requirement	250	485	22	30	137	-	-	-	-	-	Min Req
	-	-	-	-	187	-	-	-	-	-	Max. Req.

